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Research Paper

Vision-driven dynamic tunnel digital twin for real-time situation and emergency awareness

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Abstract

Effective management and safe operation of tunnel infrastructure are critically challenged by the persistent vehicle tracking across non-overlapping camera domains and the lack of real-time situational awareness for emergency incidents. This work proposes a dynamic digital twin framework to create a high-fidelity virtual representation that synchronously updates with physical tunnel environments by leveraging existing Closed-Circuit Television (CCTV) networks. The core is a novel Tunnel Re-identification Model (TRIM) that can create discriminative feature embeddings for seamless vehicle tracking and re-identification across multi-camera domains. Comparative validations unequivocally demonstrate that the TRIM significantly outperforms Convolutional Neural Networks (CNNs) and standard Vision Transformer backbones, achieving a state-of-the-art mAP@10 of 0.684. Analysis of attention heatmaps reveals its ability to generate a comprehensive and granular feature representation, focusing on preserving high-quality characteristics for specific vehicle identity. Furthermore, the emergency response ability of the digital twin is augmented with a vision-based fire detection module to real-time localize and visualize emergent tunnel fire events in the digital twin. By consolidating and distilling vision data from CCTV cameras into an intuitive representation, the proposed framework enhances situational judgment for operators and significantly reinforces the safety resilience of tunnel infrastructure without requiring extensive hardware redevelopment.

Keywords: Digital twin; Vehicle re-identification; Emergency response; Traffic management; Smart tunnel

1 Introduction

As fundamental transportation infrastructure, road tunnels are the key nodes of city operation and personnel interaction. Due to the inherently semi-enclosed environments and complicated facilities of tunnels, traditional tunnel operation and maintenance mostly rely on isolated monitoring systems and preset management plans (Liu et al., 2024). Operational data, including videos, textual reports, and binary sensor signals, commonly lags behind changes in real-world tunnels, and its presentation is not in a straightforward and easy-to-understand format. Operation and maintenance teams necessitate information-intensive distillation and analysis of heterogeneous data from tunnel sensor networks (Yin et al., 2020). Traditional tunnel operation lacks responsiveness to the dynamic changes in traffic flow and the complexity of unexpected events, with critical issues such as insufficient perception capabilities, delayed decision-making responses, and high operational costs (T. Li et al., 2024).

Digital twins, as a bridge connecting the physical world and the digital space, offer a revolutionary approach to solving the aforementioned problems and achieving smart tunnel operation (Wu et al., 2025). Its data-driven integra-

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tion with tunnel facilities and historic records enables real-time awareness and real-scene modeling of digital twin models for tunnel scenarios (Yu et al., 2021). This further facilitates traffic management and emergency response by prompt vehicle positioning and immediate data transmission, thereby providing a visual platform for data integration and judgment (Ye et al., 2023). Furthermore, the implementation of digital twins allows a unified data platform to interconnect different functional modules and subsystems, such as equipment signal controllers and fire services (X. Zhang et al., 2024b).

Despite these benefits, how to fully utilize data, perform real-time updates, and reflect changes of the real world remains the focus for tunnel digital twins (Liu et al., 2021). Current solutions have limited practicality because of the lack of cross-domain perception and the incompatibility of multi-source sensor data. This data silo degrades tunnel digital twins into low time-sensitivity systems with poor interaction between physical devices and virtual models. One of the challenges lies in persistently tracking vehicle flow inside tunnels, as well as immediately reacting to emergency incidents such as fires (Gao et al., 2021).

Given these gaps, this study proposes a dynamic tunnel digital twin framework for simultaneously updating vehicle behaviors and reflecting occurrences in accordance with the physical world. Rather than functioning as a fully automated closed-loop digital twin, the proposed framework serves as an advanced data fusion and perception layer within the comprehensive digital twin architecture (Chang et al., 2024). By employing a Transformer-based re-identification model (TRIM) to conduct visual feature extraction and matching, the constant tracking and re-identification (Re-ID) of the same vehicle from multiple camera areas can be performed within the full-domain digital twin. Along with the regular traffic monitoring function, a vision-based fire detection unit is coupled into the digital twin, which provides accurate fire coordinates and risk estimations. This dynamic tunnel digital twin is established directly on Closed-Circuit Television (CCTV) networks, meaning it can be conveniently deployed in existing and upcoming tunnels without redevelopment. By consolidating multi-camera vision data, it establishes a fundamental data-fusion platform, thereby providing the essential prerequisite data infrastructure for future intelligent decision-making and automated facility control.

2 Literature review

2.1 Digital twin in tunnel lifecycle

Over the period of tunnel construction, uncertain and changeable geological conditions will be encountered, which are gradually exposed as excavation or blasting progresses (Desai et al., 2022). Through integrating on-site sensing methods, including image recognition, oblique photography, laser scanning, and various physical sensors, digital twins are allowed to quickly and accurately access geotechnical and structural responses, thereby supporting tunnel construction (L. Zhang et al., 2024). Considering that tunnel construction involves multidisciplinary knowledge that contains mechanics, thermodynamics, geology, and materials science, digital twins present computability and predictability to simulate multi-physics coupling and avoid potential structural failures with fine-grained descriptions (Fang et al., 2024).

During the operation and maintenance phase, digital twins serve as a comprehensive and intelligent management platform, providing a holistic and real-time overview of the tunnels' performance and environment. By integrating with disparate systems, such as electromechanical equipment (Yu et al., 2023), lighting controls (Shen et al., 2022), traffic signals (Zhao et al., 2022), and structural health monitoring (Wang et al., 2025), digital twins create a unified representation of tunnel operational conditions. A practical exemplification involves designing a digital twin framework to adopt surveillance videos to reconstruct an immersive virtual tunnel and couple with tunnel facilities (Z. Wu et al., 2022). This enables operators to predict the future performance of these assets and minimize the losses of traffic accidents, moving beyond reactive problem-solving to a more predictive and optimized management paradigm (Khan, 2025). Historic traffic patterns can be superposed onto this digital twin model, allowing for the proactive adjustment of traffic management strategies at the specific time point to ensure smooth and efficient transit (Rezaei et al., 2023). Furthermore, tunnel digital twins enable the sophisticated optimization of critical subsystems. For instance, ventilation and lighting systems can be dynamically adjusted based on actual traffic volume and air quality data, thereby significantly reducing energy consumption and operational costs while adhering to safety standards (Yang et al., 2024).

Although these applications have significantly empowered smart operation and maintenance of tunnels, current studies in this field are still limited by data silos (Chang et al., 2024). They lack a unified mechanism for associating and integrating tunnel lifecycle data, which restricts continuous environmental perception of tunnel scenarios. Existing methods have difficulty enabling mutual interaction between various data, including CCTV videos, sensor measurements, and facility conditions, resulting in digital twin frameworks that fall short in supporting operational and management decisions.

2.2 Digital twin in emergency response

In the domain of emergency response and disaster mitigation, digital twins are emerging as a cornerstone for achieving new levels of efficiency, safety, and sustainability (Fan et al., 2021). This high-fidelity replica of the physical world makes key emergency information available for security staff from Internet of Things (IoT) networks, traffic systems, and environmental monitors (Huang et al., 2022). The resulting dynamic twin facilitates a holistic

understanding of the operational state, enabling a shift from reactive to predictive response. Advanced analytics and machine learning algorithms applied to digital twins can assist with flood prevention by integrating rainfall data and river water level information from sensors in flood-prone areas (Thakur, 2024). Regarding urban wildfire control measures, digital twins offer a systematic pipeline that integrates remote sensing data (Wang et al., 2024), fuel distribution (Lewis et al., 2024), meteorological information (Hyeong-su et al., 2019), and aerial field images (Shao et al., 2023) to drive complex fire spread models. This enables emergency departments to predict the spread path and speed of wildfire and its products in a virtual environment, determining the efficiency of different firefighting strategies and evaluating residential impact (Halem et al., 2024).

The application of digital twins fundamentally transforms emergency response within tunnel environments, where rapid and informed decision-making is paramount (Yu & He, 2022). In the event of an incident, such as a fire or collision, the digital twin provides immediate and comprehensive situational awareness to emergency services and tunnel operators by tracking high-fire-risk vehicles and forecasting potential fire power (X. Zhang et al., 2024b). For example, it can predict the propagation of fire and the dispersion of smoke with high accuracy, considering ventilation status and tunnel geometry through communicating with external artificial intelligence (AI) models (X. Wu et al., 2022). This predictive capability is invaluable for formulating effective response strategies, such as optimizing the activation of fire suppression systems and dynamically adjusting ventilation (Zeng, 2024). Moreover, digital twins can simulate multiple evacuation scenarios in seconds, identifying the most efficient egress paths and communicating this guidance dynamically to travelers via dynamic signage (Han et al., 2020; Wong et al., 2025).

Overall, the ability of digital twins for simulation, prediction, and guidance provides an advantage that enables responders to mitigate the negative impact, protect human life, and restore normal operations with minimized delays. However, to capture changing fire events and support time-critical emergency response, low-delay information synchronization between real tunnels and digital twins is required (Wolf et al., 2022). Most investigations have selected sensors already positioned in tunnels as data sources to implement digital twins, where situational awareness is limited compared to CCTV networks. This may cause the critical unreliability of emergency information.

2.3 Vision-based sensing in tunnels

While the aforementioned applications highlight the potential of digital twins in tunnel management and emergency response, realizing these capabilities heavily depends on real-time data acquisition and sensing ability (Liu et al., 2025). Situational awareness in digital twins has been limited by a reliance on single-point-based sensors, which can cause a critical latency between the emergency occurrence and response. To overcome this sensory bottleneck, recent research has increasingly turned towards vision-based sensing. With technical advances in computer vision, it has become increasingly mature to take full advantage of widespread CCTV camera networks to sense internal tunnel environments without extra installations of expensive LiDARs or motion detectors (Diren & Althen, 2023; Rios-Cabrera et al., 2012). Meanwhile, vision data provides more diverse information than that provided by lasers or millimeter waves, which supports the real-world modeling and awareness of tunnel scenes (Attard et al., 2018). Some investigations have demonstrated the reliability and efficiency of vision data on tunnel construction (Y. Li et al., 2024), 3D digitalization (Xue et al., 2021), defect inspection (Li et al., 2023), traffic management (Lee & Shin, 2019; Tahir et al., 2023), and emergency response (Cheng et al., 2025; X. Zhang et al., 2024a).

Despite these advancements in single-camera data analytics, a critical gap remains in the continuous tracking of dynamic entities across the entire tunnel infrastructure. A significant amount of information in the visual data captured by CCTV networks has not been used efficiently, remaining confined to processing and analysis within a single camera domain. The information isolation between cameras with non-overlapping coverage areas results in difficulties in real-time visualization and cross-region vehicle re-identification in tunnels. This limitation directly motivates the development of the digital twin framework proposed in this study, which seeks to transform disjointed camera feeds into a full-domain perception layer for the dynamic digital twin.

3 System architecture

3.1 Top-level design

The proposed dynamic digital twin adopts CCTV cameras as the data source for tunnel environment sensing, which can significantly minimize hardware dependency. As shown in Fig. 1, the digital twin framework employs a multi-stage pipeline to transform multi-source video feeds into a coherent and dynamic representation. It first detects and tracks vehicles across real-time streams. A Re-ID module then extracts vehicle features, computes similarity scores against a feature gallery, and assigns a global identifier (GID) to each vehicle. In the event of a fire, the framework rapidly localizes the fire disaster and visualizes it on the user interface, providing critical situational awareness to initiate swift and targeted emergency responses.

This dynamic tunnel digital twin consists of multi-functional submodules, which support video decoding, vehicle tracking, vehicle Re-ID, fire detection, and model rendering. Therefore, the architecture is realized through a component-based design that enhances systematic scalability. Unified Modeling Language (UML) is a useful

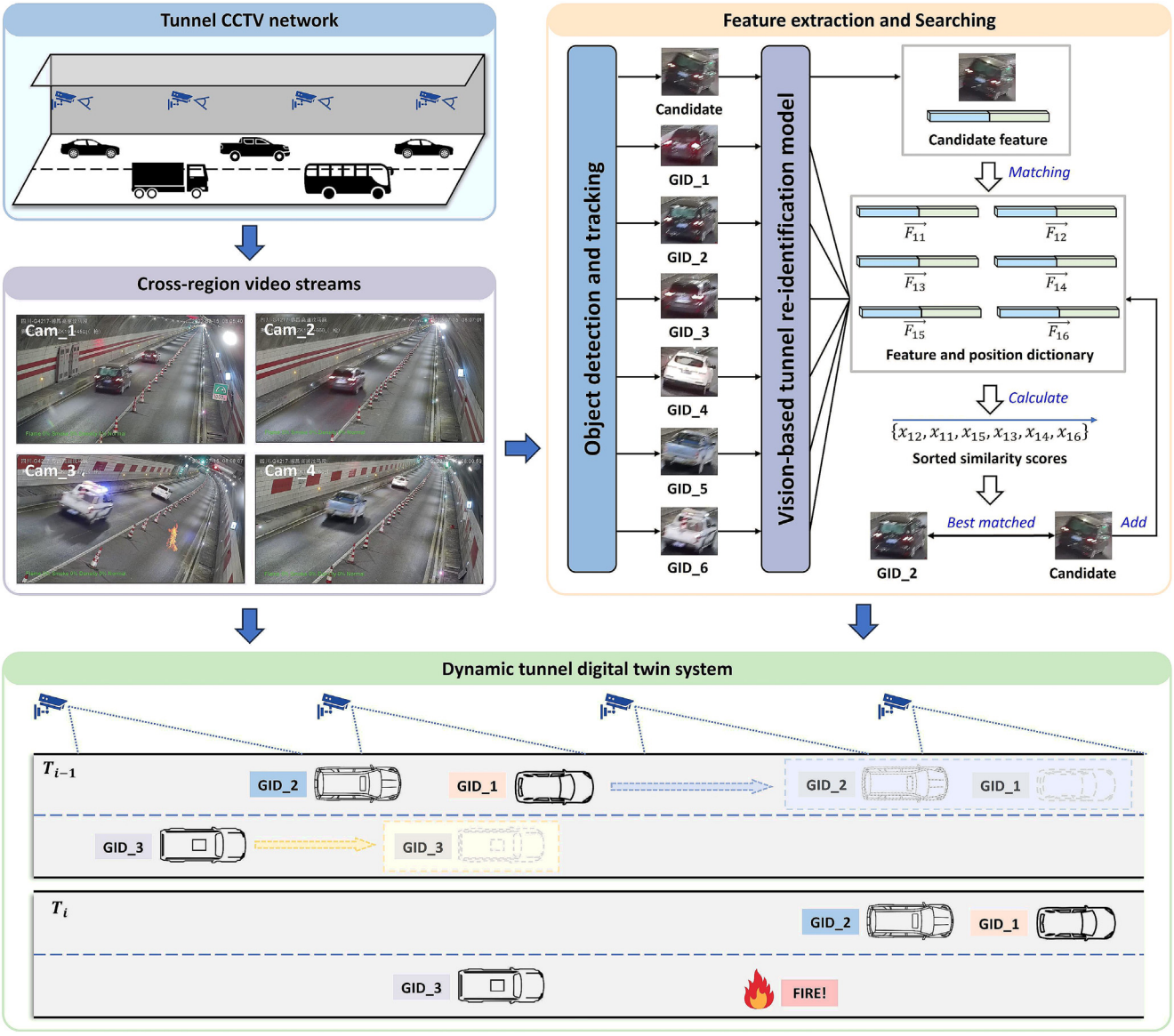


Fig. 1. Conception of the proposed dynamic tunnel digital twin framework.

way to illustrate the data interaction between various components (see Fig. 2) (Medvidovic et al., 2002). The data flow initiates with the ingestion and decoding of cross-region video streams from a network of CCTV cameras distributed throughout the tunnel infrastructure. An object detection and tracking module processes these streams to identify and assign transient local identifiers (LID) to vehicle entities within individual camera views.

The core of the digital twin is the vehicle Re-ID module, which tracks vehicles across non-overlapping camera views. It extracts a feature vector from each detected vehicle and matches it against a dynamic feature dictionary. By calculating similarity scores, the system correlates vehicle instances across different cameras to assign persistent global identifiers. Furthermore, an identity corrector integrates spatiotemporal data, such as camera layout and sequence, to validate these identifiers. This module assesses the coherence of each detected vehicle trajectory by verifying its conformity with the known camera topology and ensuring that the cross-camera transition times are physically plausible. Filtering out illogical Re-ID results maintains a high-fidelity representation of the tunnel environment and ensures the accuracy of continuous vehicle tracking.

Furthermore, the architecture supports functional modularization, exemplified by an auxiliary fire detection module. The fire detection task is performed by a fine-tuned YOLOv8 model trained for 150 epochs on 70 000 images. This integration demonstrates the framework's extensibility for concurrent task processing without bottlenecking the primary tracking modules. Crucially, the vehicle Re-ID and fire detection modules do not operate in isola-

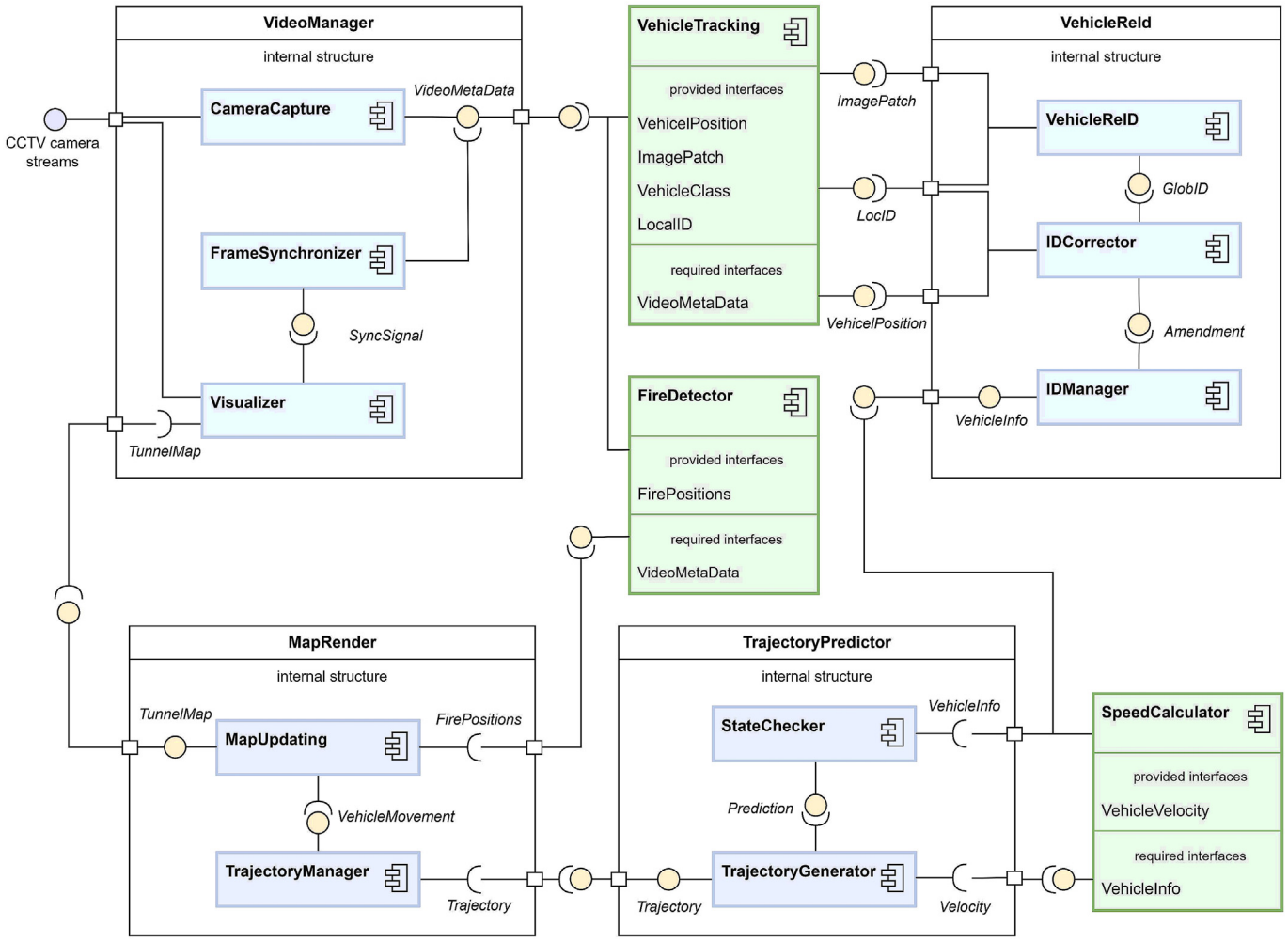


Fig. 2. Component diagram of the digital twin architecture using the UML method.

tion. They are coupled at the data fusion layer to achieve comprehensive situational awareness. While the fire detector yields spatial coordinates and event triggers, the Re-ID workflow supplies persistent vehicle identities. By mapping both data streams onto a unified spatiotemporal framework, the digital twin inherently links fire events with vehicle trajectories. Consequently, the system can instantly ascertain hazard locations alongside the quantities of trapped vehicles. Based on the reserved modular interface, this data link can further dictate the scale, priority, and urgency of subsequent emergency responses via advanced fire prediction technologies.

3.2 Vehicle Re-ID strategy

The vehicle re-identification strategy is underpinned by a deep learning architecture for extracting high-dimensional features and matching vehicle entities. While convolutional neural networks (CNNs) have been widely adopted for visual tasks, their reliance on local receptive fields makes them vulnerable to the unique challenges of tunnel environments, such as cluttered backgrounds, variable illumination, and partial occlusions. CNNs often struggle to capture the holistic structure of vehicles and are easily distracted by dominant global features like color, resulting in perceptual aliasing among vehicles with high inter-class similarity. Vision Transformer (ViT) architectures overcome these problems through their self-attention mechanism, which inherently possesses a global awareness capability (He et al., 2021). However, universal ViTs often exhibit overly sparse attention, missing fine-grained local details critical for instance-level identification. Therefore, a vision-based tunnel re-identification model (TRIM) is specifically designed based on the ViT paradigm to solve these drawbacks, as presented in Fig. 3.

Instead of processing the entire vehicle image at once, the model employs a sliding window approach to divide the image into a sequence of overlapping 2D patches. This sequence allows TRIM to convert each patch into a high-dimensional vector, with spatial relationships preserved by embedding position information into each patch. A global aggregation function is also included to summarize the entire sequence. Furthermore, a random combination strategy aggregates subsets of local features, forcing the

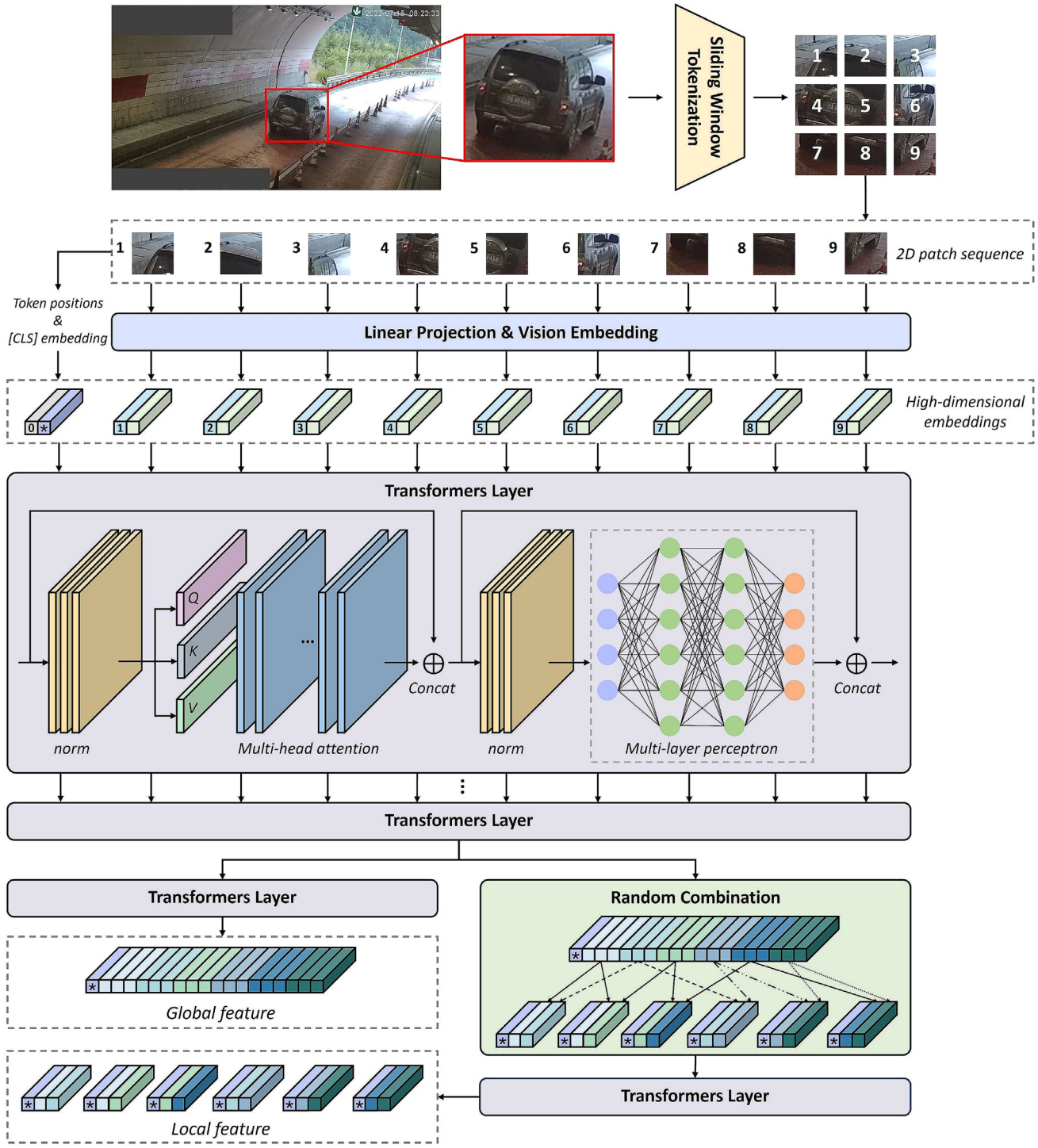


Fig. 3. Architecture of the Transformer-based TRIM feature decoder and matcher.

model to learn discriminative details rather than over-relying on a single vehicle part. This strategy can generate a multi-granularity feature representation that captures both the holistic appearance and fine-grained local details of a vehicle, ensuring reliability against the perceptual challenges inherent in tunnel environments.

Algorithm 1 details the entire procedure of this vehicle identifier assignment strategy. The feature matching procedure involves maintaining a dynamically updated global feature library and a look-up table that links vehicle features to unique global identifiers. When a vehicle is detected, its extracted feature vector is compared against the maintained library using cosine similarity. If the highest similarity score exceeds a predefined threshold, the vehicle is recognized as an existing entity, and its spatial location is updated in the look-up table. Conversely, if the score falls below the threshold, the vehicle is recorded as a newly appeared target and assigned a new identifier. This is fol-

lowed by a spatiotemporal correction module based on the camera network topology to ensure the logical consistency of the vehicle trajectory.

Algorithm 1 Main algorithm of the TRIM submodule.

Input: CCTV camera stream VidMeta
Output: Global ID of the vehicle GID, look-up table LUT

Initialization FeatLib ← {},
LUT ← {}, GID ← 1, threshold ← 0.8;
for VidFrame in VidMeta **do**
  ImgPatch, VehPos, LID, CamID = Detection(VidFrame);
  QueryFeat = TRIM(ImgPatch);
  if FeatLib is None **then**
    FeatLib[GID]= QueryFeat;
    LUT[GID]= (CamID, LID);
    return GID, LUT
  else
    for GID, VehFeat in FeatLib **do**
      SimDict← {};
      sim = dot(QeuryFeat, VehFeat)/(‖QeuryFeat‖ ∗ ‖VehFeat‖);
      SimDict[GID]= sim;
    end for
    L ← a list of all pairs (key, value) from SimDict;
    Sort L based on the values of each pair in descending order;
    bestid, bestscore = L[0];
    if bestscore < threshold **then**
      GID+= 1;
      FeatLib[GID]= QeuryFeat;
      CorrectedGID= IDCorrector (LUT, CamTopology);
      LUT[CorrectedGID]= (CamID, LID);
      return CorrectedGID, LUT
    else
      LUT[bestid]= (CamID, LID);
      return bestid, LUT
    end if
  end if
end for

4 Demonstrations and analyses

4.1 Visualization

The dynamic digital twin framework features a graphical user interface that provides operators with comprehensive situational awareness (as shown in Fig. 4). It effectively integrates raw visual data with a synthesized representation of the tunnel environment. The surveillance panel displays synchronized video streams from multiple cameras along the tunnel. These live feeds are enriched with real-time visual augmentations, including bounding boxes for vehicle tracking and immediate alerts for environmental hazards like fires. By combining direct visual evidence with machine-generated annotations, the interface allows tunnel operators to rapidly verify and assess emergency events.

A spatiotemporal replica of the tunnel environment is juxtaposed beneath the video feed, which is the core element of the digital twin. This view abstracts the complex, multi-perspective video data into a single, coherent operational map of the entire tunnel length. It plots the precise longitudinal position of every tracked entity, assigning each vehicle a persistent identifier and annotating it with dynamic attributes such as its current velocity. Crucially, this digital twin also visualizes detected events, such as the fire, at their exact location within the tunnel infrastructure.

In terms of implementation details, once the fire detection module identifies a flame within a camera's video feed, the system instantly extracts the bounding box parameters and projects these 2D pixel coordinates onto the spatiotemporal operational map of the digital twin. This transformation converts a basic visual alert into precise spatial coordinates, visualizing the exact location of the fire alongside camera positions, blind zones, and the dynamic trajectories of surrounding vehicles. By integrating this localized event data with persistent vehicle identifiers, the digital twin highlights the synergistic data linkage between the fire detection and Re-ID modules. Tunnel operators can instantly identify which specific vehicles are trapped upstream of the fire disaster. This cooperation translates isolated visual detections into a cohesive narrative, empowering operators to accurately assess the spatial relationship between the fire and trapped vehicles.

4.2 Measurement

In order to demonstrate the tracking continuity and accuracy of the TRIM module, specific metrics are included. Cosine similarity serves as a fundamental metric in feature matching, prized for its effectiveness in measuring the spatial relatedness of two vectors. This property makes it suitable for image retrieval tasks, where variations in illumination, contrast, or feature activation levels should not detract from the feature similarity (Sadbhawna et al., 2022). Within a query set $Q = \{\boldsymbol{q}_1, \boldsymbol{q}_2, \cdots, \boldsymbol{q}_m\}$, the cosine similarity can be derived for each vehicle search query $\boldsymbol{q}_i$ as:

$$\mathrm{sim}(\boldsymbol{q}_i, \boldsymbol{g}_i) = \frac{\boldsymbol{q}_i \cdot \boldsymbol{g}_i}{\|\boldsymbol{q}_i\| \|\boldsymbol{g}_i\|}, \tag{1}$$

where g_i is the vehicle feature in the retrieval gallery set $G = \{\boldsymbol{g}_1, \boldsymbol{g}_2, \cdots, \boldsymbol{g}_n\}$.

However, the cosine similarity is an absolute metric that lacks a global context regarding other candidates in the gallery. Therefore, a normalized probability score (NPS) is introduced to represent the fidelity of the candidate vehicle to the reference. By applying a variant of the softmax function, the NPS metric amplifies the separation between



Fig. 4. Visualization interface of the dynamic tunnel digital twin framework (see Video S1 and Video S2 in the supplementary materials for more details).

strong and weak matches. This provides an intuitive, global view of the searching performance and sharpens the decision boundary, which is essential for applications requiring high-confidence vehicle identification. Given a query request $\boldsymbol{q}_i$, NPS can be mathematically expressed for each retrieval object $\boldsymbol{g}_i$ as follows:

$$\mathrm{NPS}(\boldsymbol{q}_i, \boldsymbol{g}_i) = \frac{e^{\mathrm{sim}(\boldsymbol{q}_i, \boldsymbol{g}_i)/\tau}}{\sum_{j=1}^{n} e^{\mathrm{sim}(\boldsymbol{q}_i, \boldsymbol{g}_i)/\tau}}, \tag{2}$$

where τ is the temperature parameter and is set to 0.05 in this case for a better comparative effect.

To evaluate multi-round query performance and retrieval ranking quality, the cumulative matching characteristics (CMC) and precision-based metrics are adopted. The CMC metric at the top-k position (CMC@k) represents the probability of finding the correct vehicle identity within the top-k retrieved matches. The formulation for CMC@k over a given query set can be expressed as:

$$\mathrm{CMC@}k = \frac{1}{|Q|} \sum_{i=1}^{|Q|} \delta(\boldsymbol{q}_i, k), \tag{3}$$

where $\delta(\boldsymbol{q}_i, k)$ is an indicator function that equals one if the true match for the query is present in the top-k candidates, and zero otherwise.

In addition to the hit probability of the CMC metric, the precision score at the top-k position (P@k) demonstrates the proportion of correct matches within the top-k search results for a single round. The P@k score for the query $\boldsymbol{q}_i$ is defined as:

$$\mathrm{P@}k(\boldsymbol{q}_i) = \frac{1}{k} \sum_{n=1}^{k} R(\boldsymbol{r}_n), \tag{4}$$

$$R(\boldsymbol{r}_n) = \begin{cases} 1, & \text{if } \boldsymbol{r}_n \in T, \\ 0, & \text{otherwise}, \end{cases} \tag{5}$$

where $R(\boldsymbol{r}_n)$ determines whether the retrieved item $\boldsymbol{r}_n$ belongs to the ground-truth set T of all relevant items.

To account for ranking quality, the average precision at the top-k position (AP@k) is calculated, heavily penalizing models that place correct items at lower ranks. The AP@k is the average of the P@k values over all relevant items found within the top-k positions. Formally, it can be expressed as

$$\mathrm{AP@}k(\boldsymbol{q}_i) = \frac{1}{|T|} \sum_{n=1}^{k} \mathrm{P@}n(\boldsymbol{q}_i) \cdot R(\boldsymbol{r}_n). \tag{6}$$

Finally, the mean average precision (mAP@k) summarizes the AP@k scores across all query rounds. Consequently, mAP@k serves as the conclusive metric for benchmarking different models, reflecting their generalizability and average efficacy across the entire target domain rather than on isolated query tasks. It can be expressed as

$$\mathrm{mAP@}k = \frac{1}{|Q|} \sum_{i=1}^{|Q|} \mathrm{AP@}k(\boldsymbol{q}_i). \tag{7}$$

4.3 Quantitative analysis

The quantitative performance of the TRIM is evaluated through feature-level similarity analysis and practical instance retrieval tasks. A feature confusion matrix provides insights into the discriminative power of the TRIM for the vehicle feature embeddings (see Fig. 5). Vehicle features are extracted and vectorized from images by the TRIM to calculate the cosine similarity score. This matrix visualizes pairwise similarity scores between vehicles captured by three non-overlapping cameras.

A critical observation is that the intra-class similarity scores are consistently and significantly higher than the inter-class similarity scores for different vehicles when comparing images of the same vehicle across different camera views. For instance, the cross-camera similarity for vehicle GID 2 remains exceptionally high (e.g., a score of 0.957 between Cam 1 and Cam 2, and 0.942 between Cam 2 and Cam 3), whereas its similarity to other vehicles is markedly lower. This pattern is consistent across all evaluated vehicle identities, demonstrating that the TRIM model successfully generates a feature representation that is both invariant to the substantial viewpoint and illumination shifts between cameras and highly specific to each vehicle. This confirms its capacity to mitigate feature confusion and form a reliable basis for identity persistence.

A retrieval dataset of 30 images across six non-overlapping cameras (comprising 9 vehicle instances) was tested against a query set of four specific vehicles (GID 1 to 4) to evaluate the efficacy of the TRIM component. The gallery included all multi-camera appearances of these queried vehicles and interference options. As shown in Fig. 6, the NPS scores confirmed satisfactory rank-1 identification accuracy of TRIM. For GID 1 and GID 2, the model successfully assigned the correct targets with an NPS above 0.7 in each camera view, while the scores of distractors remained negligibly low to minimize false matching. However, Fig. 6(c) illustrates an inherent challenge

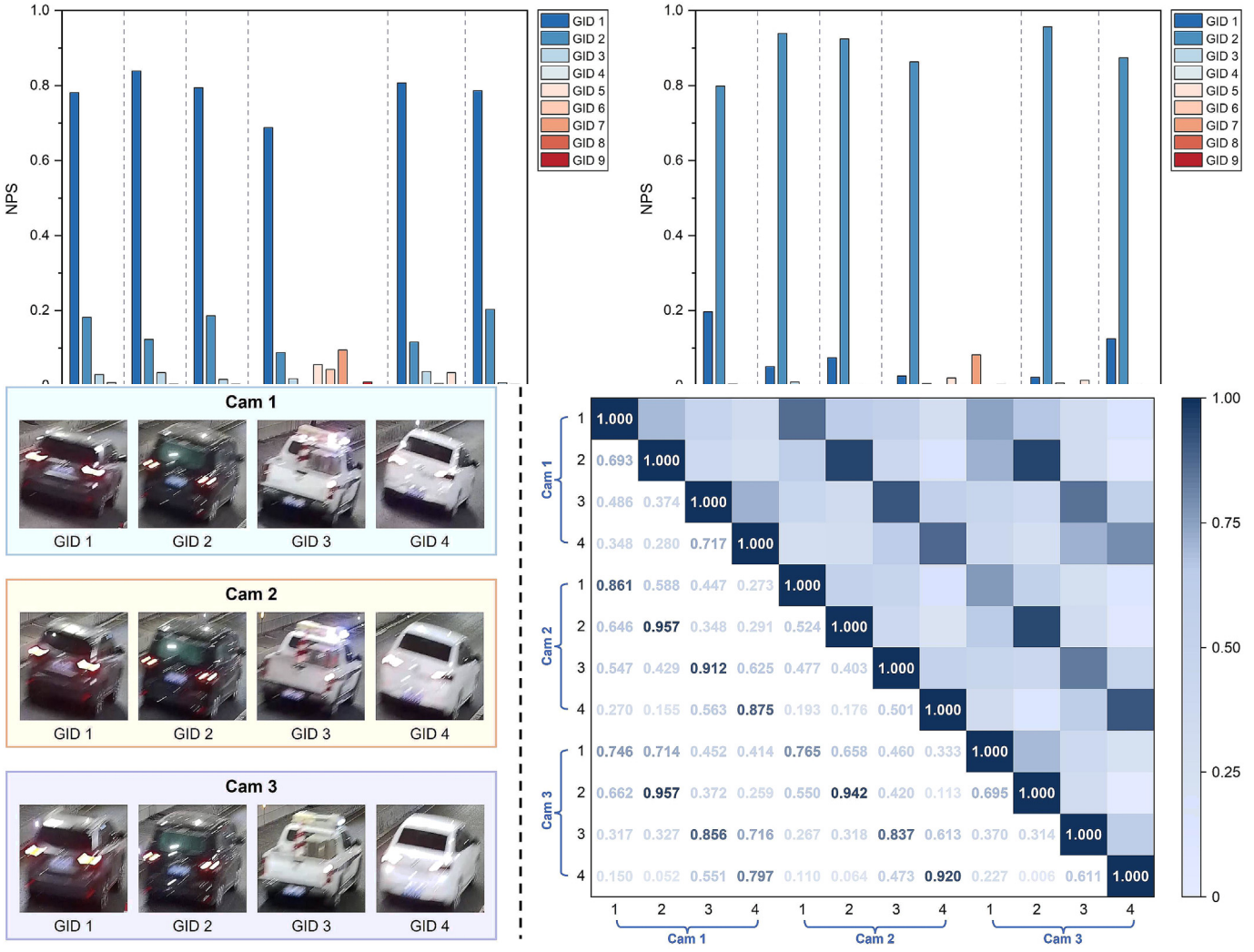


Fig. 5. Similarity confusion matrix of vehicle features under three non-overlapping cameras using TRIM.

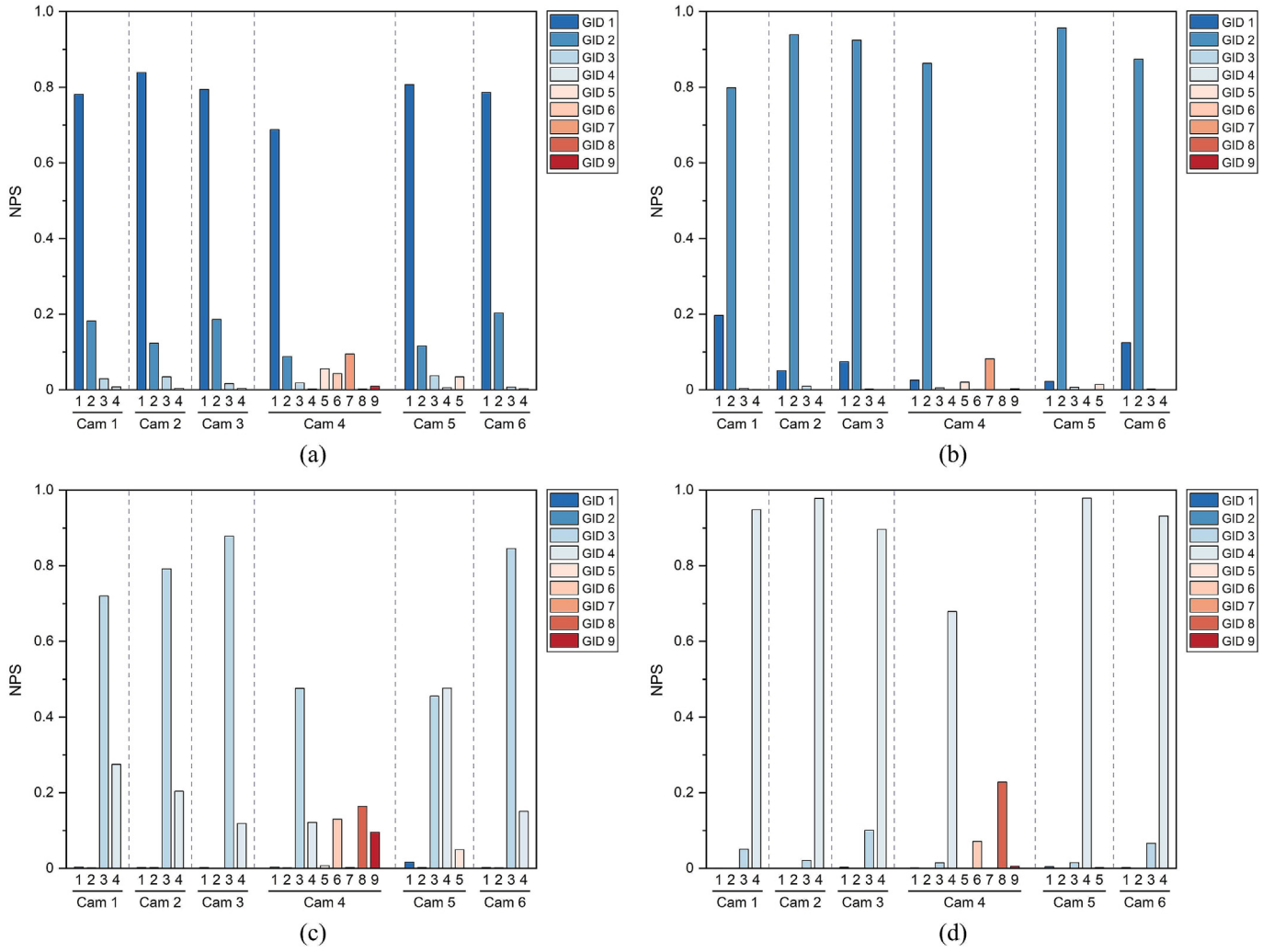


Fig. 6. Query results of specific vehicles from the vehicle gallery by TRIM. (a) GID 1, (b) GID 2, (c) GID 3, and (d) GID 4.

that GID 3 was mismatched with GID 4 in Cam 5. They are visually similar light-colored cars, which slightly reduced the discriminative margin. Despite this isolated mismatch, the consistent top-ranked success across most vehicle queries validates that TRIM delivers a credible retrieval capability for persistent vehicle tracking within the dynamic digital twin.

4.4 Comparative validations

To qualitatively validate the efficacy of the feature extraction strategy employed by the TRIM model, a comparative analysis of its feature extraction mechanism was conducted against two prominent backbone architectures, namely a CNN-based ResNet-152 (He et al., 2016) and a standard Vision Transformer (ViT) (Dosovitskiy et al., 2020). The gradient-weighted class activation mapping (Grad-CAM) algorithm was employed to visualize the attention heatmaps generated by each model for a diverse set of vehicle images (Selvaraju et al., 2017). It is possible to discern where each architecture focuses its attention to extract discriminative cues for vehicle re-identification (see Fig. 7). The results can clearly illustrate the superior and more task-appropriate feature learning of the TRIM model.

The ResNet-152 baseline produced coarse and diffuse attention maps that frequently spilled into irrelevant background areas, such as roads or tunnel walls. While capable of general vehicle detection, the model struggles to isolate the subject from environmental clutters, which severely compromises its effectiveness of instance-level feature identification. In contrast, the standard ViT suppressed background noise and confined its attention to vehicle instances. However, its focus was overly sparse, concentrating only on high-contrast scattered patches. Although it successfully pinpoints discriminative regions, it fails to capture the vehicle's holistic appearance. This reliance on a few discrete features renders the ViT model highly vulnerable to occlusion, motion blur, and challenging lighting conditions.

The attention mechanism of TRIM synthesizes the strengths of the preceding models while mitigating their

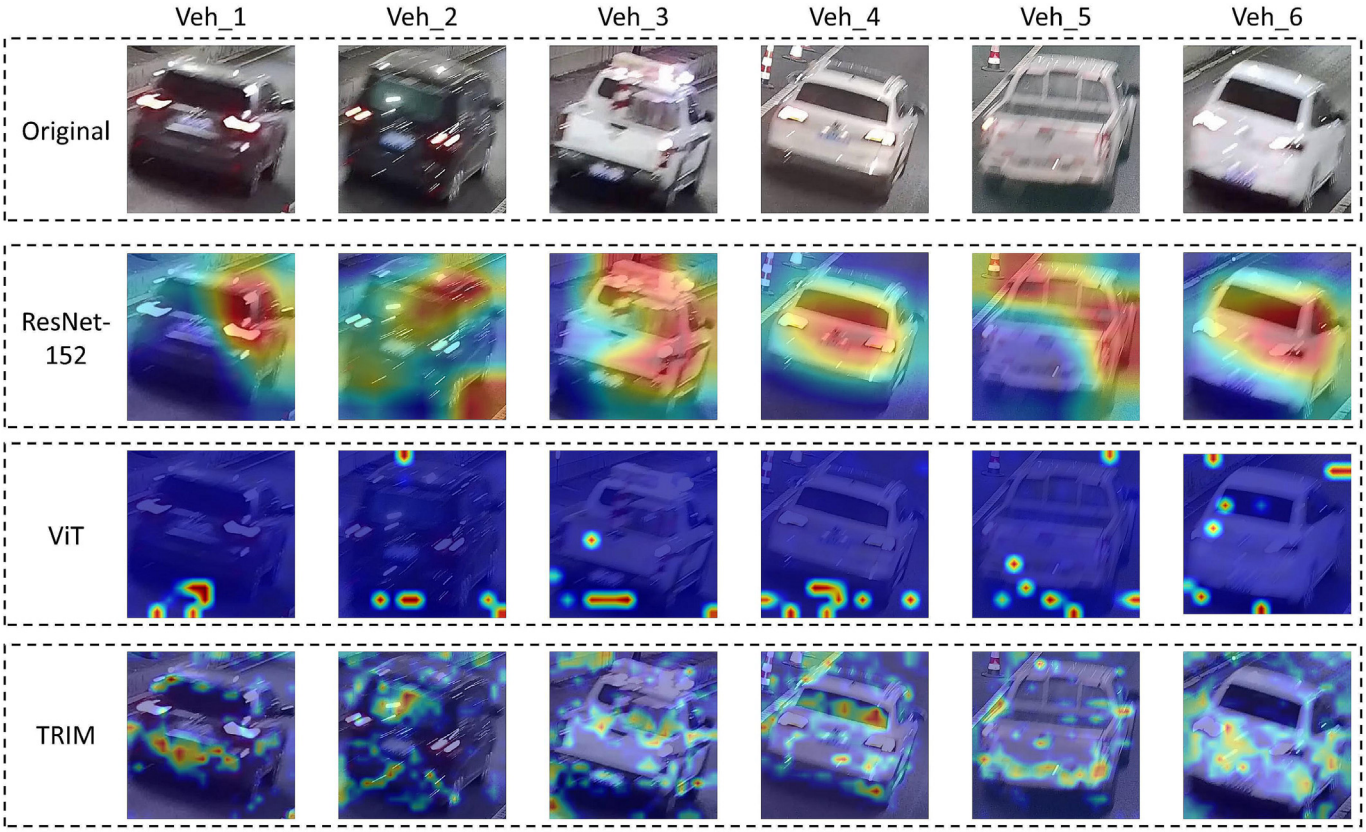


Fig. 7. Feature attention heatmaps of the candidate backbones using Grad-CAM.

respective weaknesses. The heatmaps reveal that TRIM learned a feature representation that was simultaneously comprehensive and granular. The attention field accurately conformed to the vehicle's overall contour, effectively segmenting it from the background, yet it also exhibited dense and high-intensity activations across a distributed set of detailed local regions. This included taillight clusters, window contours, rooflines, and other unique structural elements. This visualization implies that the TRIM model is adept at capturing a rich, multi-scale set of preserving identity features across the entire object. This holistic and detailed attention strategy explains the enhanced performance of TRIM in reliably reidentifying vehicles under the challenging and variable conditions of a real-world tunnel environment.

Vehicle retrieval experiments were conducted for the ResNet-152 and ViT models using the same query and retrieval datasets as those for TRIM. Figure 8 illustrates the performance of the ResNet-152 model, where the network achieved satisfactory discrimination for GIDs 1, 3, and 4. In these cases, the NPS scores were consistently highest for the correct vehicle instances across most camera views, indicating that the CNN-based features are sufficient for these distinct vehicle identities. However, a significant failure was observed in Fig. 8(b) regarding GID 2. The model exhibited substantial confusion, erroneously assigning high NPS scores to GID 1 in many camera views (indicated by the light blue bars) rather than the correct target. It is noticeable that GID 1 and GID 2 are both dark-colored vehicles. This suggests that the convolution-based architecture may overfit specific local features and struggle to resolve the fine-grained visual differences between GID 1 and GID 2, leading to frequent misclassification.

As shown in Fig. 9, the ViT model offers an improvement over the ResNet-152 baseline but falls short of the performance established by the TRIM architecture compared to Fig. 6. While the ViT model successfully corrected the misclassification of GID 2, as seen in the CNN-based results, and identified the correct vehicle in every view, it failed to obtain considerably lower confidence margins than TRIM. The NPS peaks in Fig. 9 typically hovered around 0.2 to 0.6, whereas those that are frequently observed in Fig. 6 exceed 0.8. Besides these, ViT displays occasional ambiguity, evidenced by the probability scores assigned to the incorrect identity alongside the correct target, such as querying GID 1 within the view of Cam 1. This comparison indicates that while standard Vision Transformers improve global feature capture, the proposed TRIM model provides a superior synthesis of features, resulting in a more decisive and accurate retrieval system.

An interesting observation is that the CNN-based model (ResNet-152) retrieved the correct vehicle for GID 3 in Cam 5, while Transformer-based models (TRIM and ViT) both failed. This is considered a reflection of the fun-

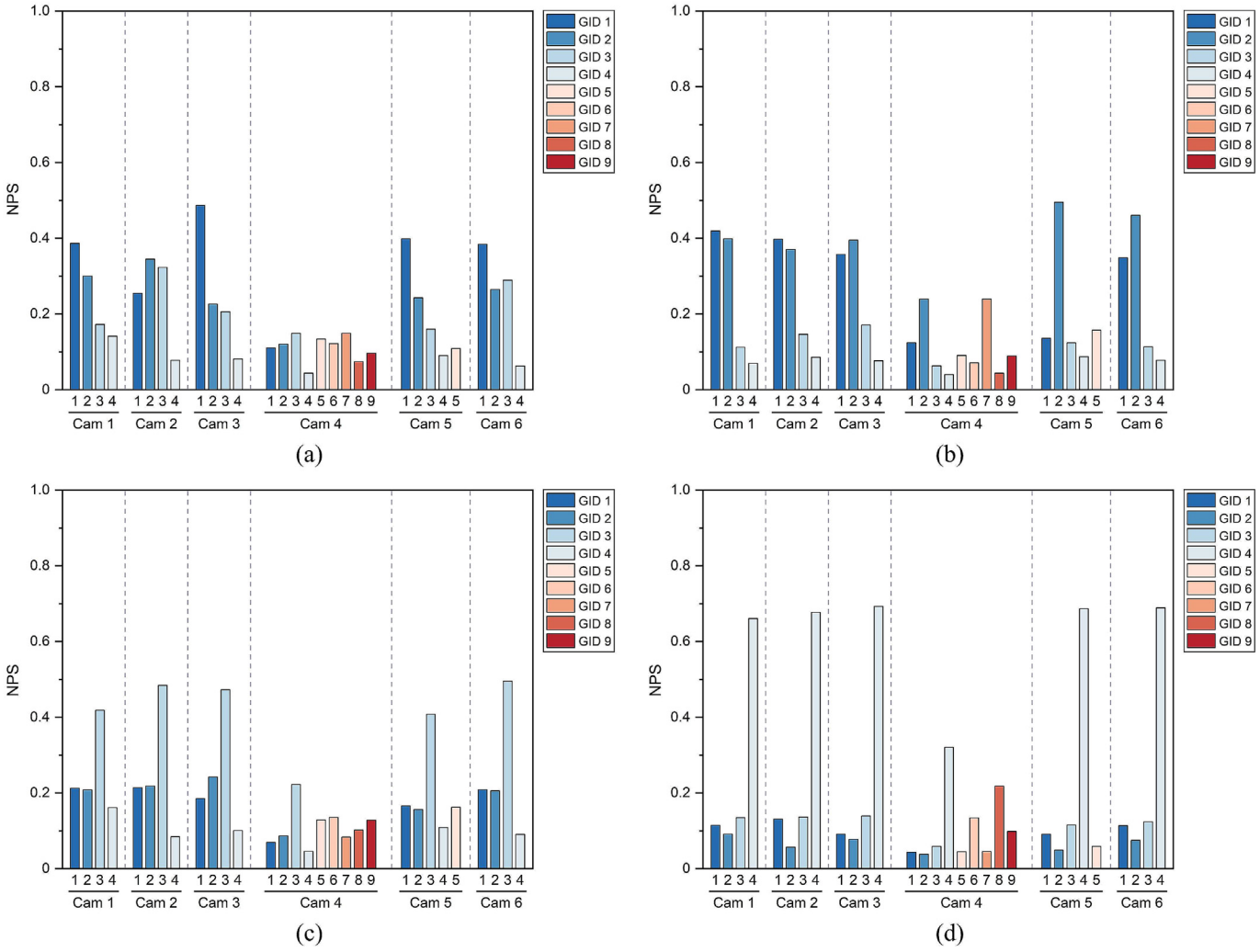


Fig. 8. Query results of specific vehicles from the vehicle gallery by ResNet-152. (a) GID 1, (b) GID 2, (c) GID 3, and (d) GID 4.

damental difference in feature attention patterns. Transformer-based models prefer global attention, meaning that they analyze the entire image at once to understand the relationship between patches. In this specific case of Cam 5, the specific feature of GID 3 might be subtle (e.g., a specific rim or sticker). Transformer models likely over-smooth these small details in favor of the overall shape or background, causing them to confidently misclassify this vehicle (like GID 5). Conversely, ResNet-152 is a convolution-based architecture designed with a local inductive bias. It preferentially extracts high-frequency local features (such as edges and textures) before global understanding of the image. This allows CNN-based models to capture a subtle but critical feature of GID 3, thereby correctly identifying the target.

4.5 Benchmark testing

To further validate the generalization of the proposed TRIM model and benchmark its performance against contemporary task-specific models, an extensive evaluation was conducted on the widely adopted VeRi-776 public dataset (Liu et al., 2018). This large-scale dataset is specifically designed for vehicle retrieval tasks in surveillance environments, including high intra-class vehicle differences and severe variations in illumination, viewpoint, and occlusion. To be directly compared against task-specific Re-ID models rather than generic visual backbones, two different architectures were introduced as additional baselines. The Attention Generalized Mean Pooling with Weighted Triplet Loss (AGW) baseline is used to leverage advanced feature aggregation in metric-learning methods (Ye et al., 2021). BoTNet is a network that synthesizes the local inductive biases of convolutions with the global self-attention of Transformer structures, which is selected as the baseline of hybrid CNN-Transformer methods (Srinivas et al., 2021).

As detailed in Table 1, the quantitative performance was measured using CMC metrics at ranks 1, 5, and 10, alongside mAP metrics at various cut-off positions. The CNN-based ResNet-152 performs moderately better (CMC@1 of 0.291); however, it struggles to overcome the perceptual

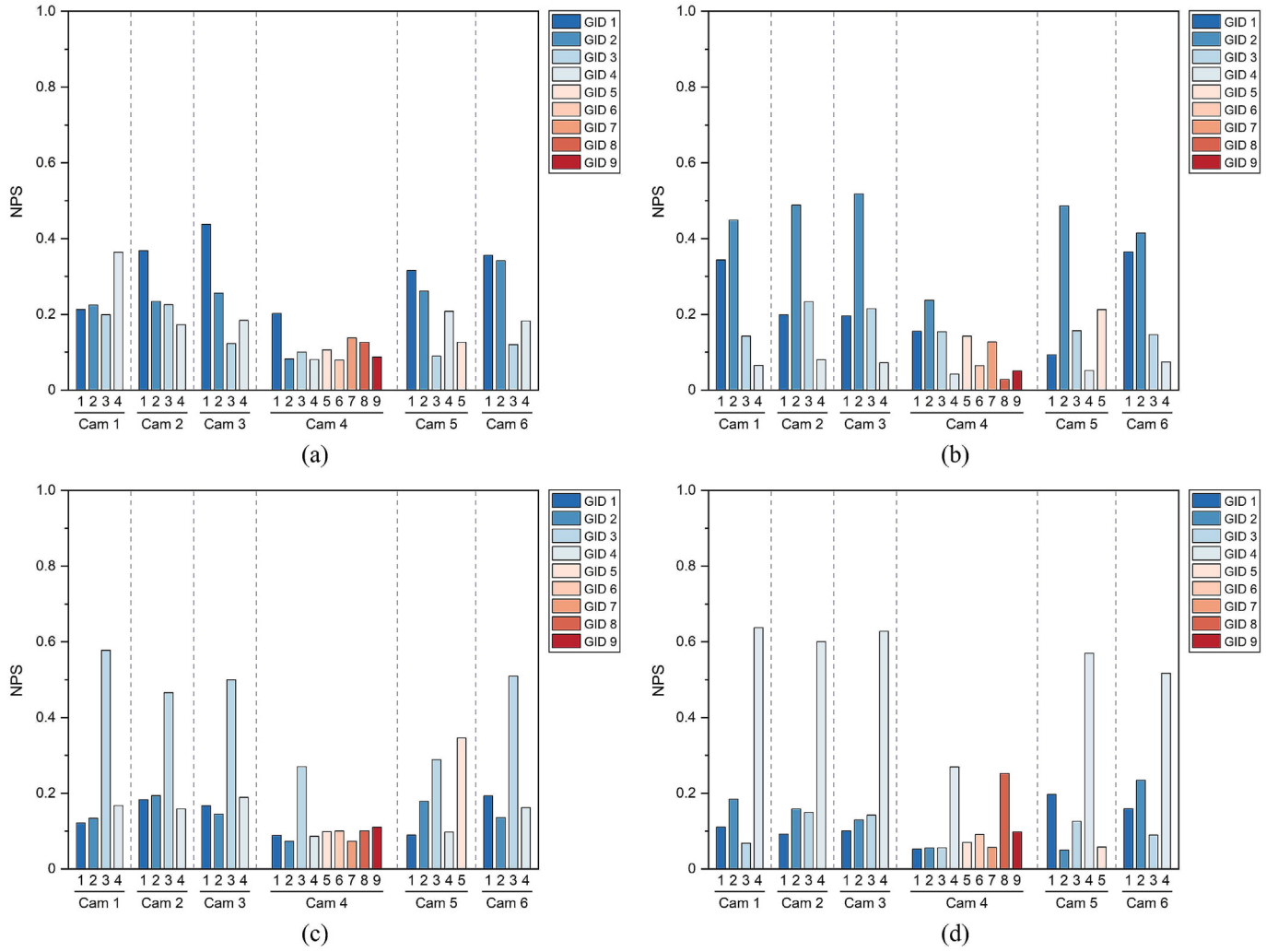


Fig. 9. Query results of specific vehicles from the vehicle gallery by ViT. (a) GID 1, (b) GID 2, (c) GID 3, and (d) GID 4.

Table 1
Complexity and query precision of the candidate models on the vehicle dataset.

Models	Params (10^6)	GFLOPs	CMC@1	CMC@5	CMC@10	mAP@10	mAP@50	mAP@100	mAP@all
ResNet-152	58.14	11.56	0.291	0.484	0.585	0.335	0.262	0.221	0.127
ViT	86.39	18.30	0.190	0.378	0.495	0.245	0.197	0.166	0.091
AGW	25.08	4.08	0.295	0.495	0.594	0.347	0.276	0.238	0.127
BoTNet	10.44	3.43	0.334	0.533	0.619	0.375	0.296	0.250	0.130
TRIM	151.49	89.24	0.705	0.855	0.894	0.684	0.549	0.499	0.444

aliasing caused by the high inter-class similarities typical in vehicle appearances. The standard ViT exhibits the lowest performance (CMC@1 of 0.190, mAP@10 of 0.245). This is largely attributed to the lack of translation equivariance and local inductive bias, making it prone to overfitting or failing to capture structural vehicle semantics. The integration of specialized metric-learning-based strategies demonstrates measurable benefits. The AGW model slightly outperforms the generic ResNet-152 model, illustrating that a designed triplet loss space can enhance feature separability. By synthesizing local feature extraction of CNNs with the global self-attention of Transformers, BoTNet can capture a more balanced representation, with a CMC@1 of 0.334 and an mAP@10 of 0.375.

While these task-specific and hybrid-structure models yielded incremental improvements over the generic vision models, they still displayed limited discriminative power. In contrast, the proposed TRIM model significantly outperforms all evaluated baselines by a substantial margin, achieving a state-of-the-art CMC@1 of 0.705 and an mAP@10 of 0.684. This performance superiority is driven by its multi-granularity feature extraction. Instead of directly using generated feature representations, the random combination of feature vectors allows TRIM to simul-

taneously capture both the holistic appearance and fine-grained local details of a vehicle. This strategy prevents the model from relying solely on global features (e.g., vehicle color or main contours) and enforces the recognition of discriminative local details (e.g., stickers or distinct taillight clusters).

The superior discriminative ability of TRIM comes with an increased computational budget, requiring 151.49 million parameters and 89.24 giga floating-point operations (GFLOPs). This increased computational overhead is necessitated by the multi-granularity self-attention mechanism, which conducts dense matrix multiplications to construct a holistic and granular feature representation. However, the architectural cost yields a remarkable 104% relative improvement over ResNet-152 in the mAP@10 metric. While it is a deliberate trade-off prioritizing retrieval accuracy, the practical feasibility of deploying the TRIM model without bottlenecking the real-time digital twin is explicitly verified in the subsequent section.

5 Discussions

5.1 System efficiency

To rigorously substantiate the real-time and dynamic capabilities of the proposed digital twin framework, a comprehensive end-to-end latency benchmark was conducted under high-load conditions. As shown in Table 2, the system was stress-tested by concurrently processing synchronized video streams from 19 non-overlapping CCTV cameras. During the evaluation, the tunnel environment maintained a dense traffic flow, yielding an average of 228 vehicles per system processing step.

The end-to-end pipeline latency is primarily distributed across four modular stages: vehicle detection, parallel vehicle tracking, TRIM-based Re-ID, and global map rendering. The vehicle batch detection stage demonstrated its efficiency with a mean latency of 78.60 ms. The parallel tracking and the core TRIM-based Re-ID modules recorded mean processing times of 133.11 ms and 145.36 ms, respectively. The 95th percentile (P95) and 99th percentile (P99) metrics indicate that even under localized computational spikes caused by sudden increases in vehicle entities, the TRIM module consistently completes feature extraction and dictionary matching within 372.43 ms. The rendering of the digital twin interface incurs a negligible computational overhead within 1.19 ms.

Overall, the end-to-end total latency yielded a mean of 413.50 ms and a P95 of 651.95 ms. This means that the framework can still maintain a sub-second refresh interval even under high-density traffic conditions and across multiple high-resolution video streams. Despite the high model complexity of the TRIM module established in the previous benchmark testing, the component-based design effectively mitigates potential data congestion. The computational overhead of the Re-ID module does not translate into an operational delay. It ensures that the digital twin promptly reflects physical traffic flow and emergency events without perceptible operational lag, thereby fulfilling the real-time requirements for human-in-the-loop emergency response.

5.2 Implications for smart tunnel operation

While this perception pipeline establishes the physical-to-virtual data flow, the mature architecture relies on a bidirectional feedback loop. The analytical conclusions derived within the virtual digital twin, such as vehicle positions and early fire coordinates, serve as immediate trigger inputs for physical space actuation. This virtual-to-physical feedback will guide physical tunnel infrastructure, enabling automated facility adjustments and informed human-in-the-loop interventions. As shown in Fig. 10, the convergence of vision-based dynamic digital twins with technologies centered on human interaction and emergency response presents profound implications for enhancing tunnel safety operation and risk resilience.

The fusion with the Artificial Intelligence of Things (AIoT) creates a symbiotic loop between the physical tunnel and its virtual counterpart, enabling high-fidelity monitoring and automated control (Xie et al., 2025). Concurrently, the incorporation of patrol robots, guided by the digital twin, can facilitate automated inspections of structural integrity and equipment status, mitigating the expense associated with human-led inspections and tunnel reconstructions (Deng et al., 2026). The integration of an Intelligent Dynamic Exit Sign (IDES) system can guide occupants to the safest egress path by altering signage in real-time according to on-site situations (Wong et al., 2025). It is a critical factor in minimizing evacuation time and potential casualties. Large language models (LLMs) provide a natural language interface for interacting with tunnel sensor data, allowing actionable insights for streamlining maintenance and operational processes

Table 2
End-to-end latency of the dynamic digital twin framework under concurrent stress testing.

Processing stage	Mean (ms)	Median (ms)	P95 (ms)	P99 (ms)
Batch vehicle detection	78.60	78.33	81.77	81.95
Parallel vehicle tracking	133.11	158.90	209.64	210.10
Vehicle Re-ID	145.36	101.99	355.21	372.43
Digital twin map rendering	1.19	0.75	2.37	2.59
Total latency	413.50	430.85	651.95	657.12

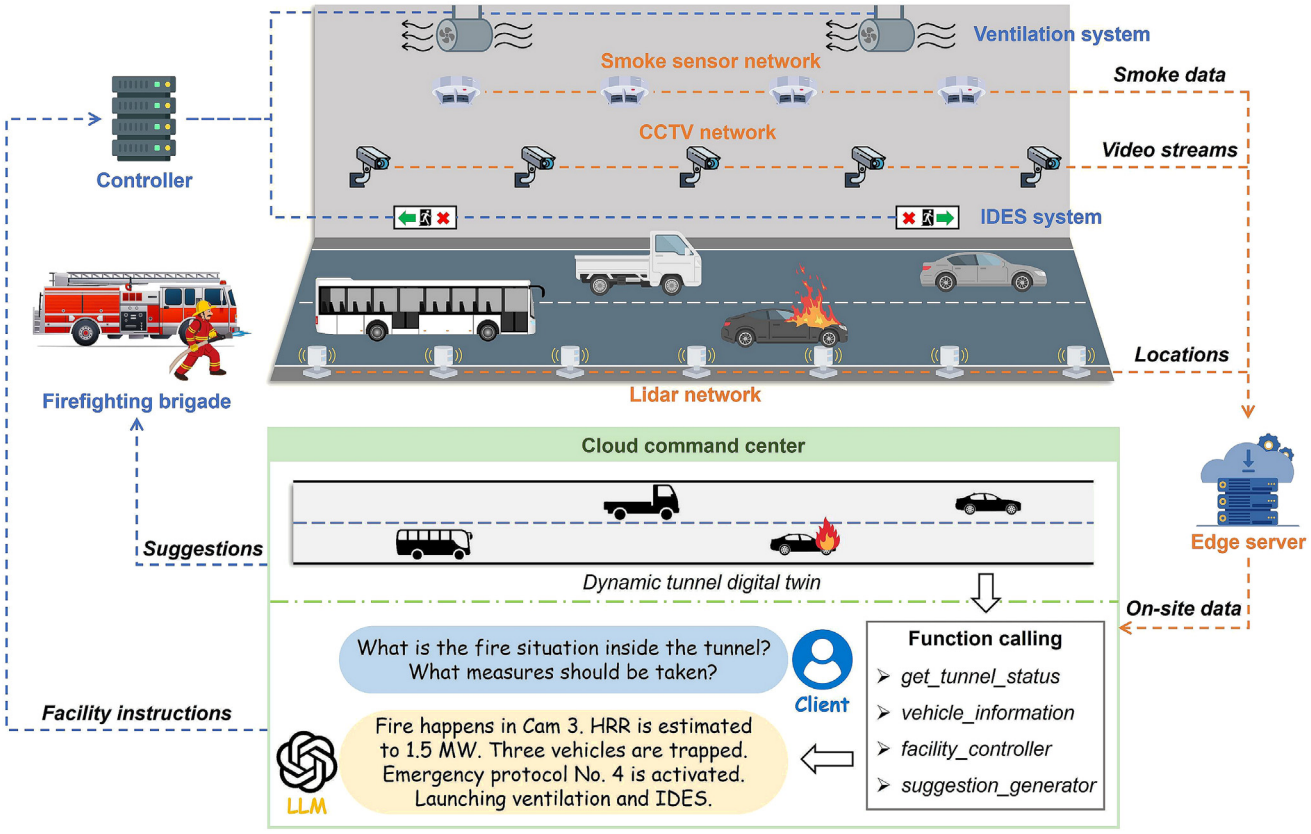


Fig. 10. Envisioned coordination with tunnel facilities to achieve intelligent tunnel operation and fast emergency response.

(Otal et al., 2024). This effectively realizes the closed-loop feedback between the virtual digital twin and physical tunnel facilities. The digital twin aggregates physical perception data into virtual conclusions, and LLMs subsequently translate them into physical execution commands (Xie et al., 2026). These integrated approaches also include a shift from reactive to predictive maintenance strategies, such as optimized energy consumption through intelligent ventilation and lighting control, improving the lifecycle management of tunnel infrastructure.

The primary engineering design of this vision-driven framework is rooted in proactive early intervention. While purely vision-based systems inevitably face severe degradation under extreme late-stage fire conditions, such as dense smoke or complete power outage, this digital twin is designed to capitalize on the critical early time window of a fire incident. By providing sub-second situational awareness during the incipient stages of a fire, the framework empowers operators to rapidly execute firefighting protocols and activate ventilation systems before severe environmental degradation occurs. This will actively prevent the tunnel from deteriorating into an extreme scenario.

5.3 Limitations and future work

The current implementation primarily solves the challenges of situational perception and data fusion of tunnel environments, which are the underlayer of a fully operational digital twin system. The focus of future research will remain on how to achieve subsequent upper-level intelligent decision-making and closed-loop facility control based on the proposed framework. This involves establishing a bidirectional interaction mechanism. Digital twins will not only sense the physical world in real-time but also command infrastructures based on the decisions made within the digital space. This transition will ultimately evolve the framework from a passive perception platform into an active, automated management agent.

Although the vision-based vehicle re-identification approach designed by this work has proven its efficiency and reliability, it may be affected by visual features of vehicles, particularly in handling high inter-class similarities. Figure 11 provides a similarity matrix across six distinct vehicles that substantiates this challenge. For example, two different white vehicles, Veh_2 and Veh_4, exhibit a high similarity score of 0.753. This demonstrates that vision-based models are still limited by over-reliance on dominant, non-unique global features like color and vehicle shape, leading to perceptual aliasing in the feature embedding space. This issue is further compounded by high intra-class variation (e.g., the same vehicle appears different due to viewpoint, illumination, or occlusion), motion blur, and inappropriate installations of cameras.

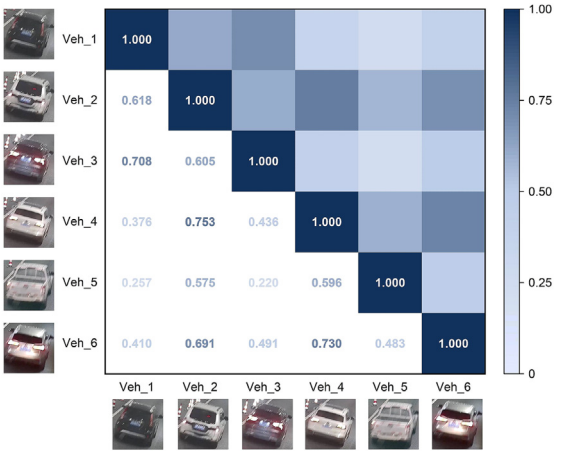


Fig. 11. Similarity confusion matrix of six different vehicles.

The current spatiotemporal identity correction module relies on heuristic assumptions regarding camera topology and expected transition times. While highly effective under normal traffic conditions, this purely vision-based approach exhibits inherent physical limitations and failure cases in high-density traffic and extreme fire scenarios. For instance, when vehicles break down or stop within a camera blind zone, or during severe traffic congestion, the actual inter-camera transition time can vastly exceed the algorithm's permissible temporal window. Under these anomalous cases, the failure of visual sensing and heuristic topological assumptions inevitably leads to ID switches or target loss.

To fundamentally overcome these limitations, a crucial advancement is to transition towards multi-source data fusion of external perceptive devices, especially LiDARs and millimeter-wave radars. This combination is envisioned to provide continuous, absolute spatiotemporal coordinates. Unlike purely visual tracking, these data modalities are immune to illumination variations and visual occlusions. By fusing this high-precision physical data with visual vehicle features, the system enables the re-ranking of the visual matching list to prioritize spatiotemporally most-probable matches. This multi-modal synergy will ensure stable identity persistence and high-fidelity situational awareness even under extreme traffic anomalies or heavy congestion.

Additionally, while the YOLO-based fire detection module demonstrates the extensibility of the proposed framework, deploying a reliable fire detection system in real-world tunnels requires rigorous quantification. It is difficult to accurately define system performance and sensitivity using open-source static datasets due to complex tunnel environments. Therefore, a full-scale fire experiment is currently being conducted in a realistic physical tunnel. The quantitative results, boundary definition, false-alarm resistance (e.g., distinguishing taillights from flames), and fire detection sensitivity of this specific vision-based module will be systematically reported in a dedicated follow-up study.

6 Conclusions

This study introduced a dynamic digital twin framework designed to enhance the operational management and safety of tunnels by leveraging existing CCTV networks. Quantitative evaluations clearly validated the superior performance of the TRIM, with an mAP@10 of 0.684, outperforming 0.335 for a CNN structure, 0.245 for a standard Transformer structure, and 0.375 for a hybrid structure. The TRIM model demonstrates a satisfactory ability to generate highly discriminative feature embeddings, which remain invariant to shifts in viewpoint and illumination while being highly specific to individual vehicles. This capability is proven through retrieval experiments where TRIM achieved the highest NPS result for correct vehicles among nearly all camera views, which indicates the consistency of a specific vehicle instance.

A comparative analysis of attention heatmaps reveals the mechanism underpinning this success. Unlike the diffuse focus of the CNN model or the overly sparse attention of the standard ViT, TRIM presents a feature representation that is simultaneously comprehensive and granular. This feature extraction strategy directly translates to reliable re-identification, even in challenging scenarios involving visually similar vehicles where other models fail. By successfully integrating high-fidelity tracking and emergent event detection, the proposed digital twin system provides an intuitive operational platform that enhances situational judgment for tunnel operators without requiring extensive hardware reconstruction.

The outlook lies in evolving this vision-driven digital twin system into a comprehensive perception and control framework. Data from heterogeneous environmental monitors is envisioned to be combined with this framework for more unambiguous situational awareness. This integration will advance the system from passive monitoring to active closed-loop control. By establishing bidirectional data flows with tunnel facilities, the digital twin will gain the capability to intelligently adjust tunnel facilities, such as autonomously activating ventilation systems, dynamically managing lighting systems, and coordinating exit signs. It enables a truly proactive paradigm for smart operation and automated emergency response in tunnel infrastructure.

Data availability

Data are made available on https://github.com/drong37/dynamic_tunnel_digital_twin.

CRediT authorship contribution statement

Rong Deng: Writing – original draft, Visualization, Software, Methodology, Investigation. **Siqi Zhu:** Validation, Software, Formal analysis. **Tong Lu:** Validation, Investiga-

tion. **Wei Wang:** Investigation, Data curation. **Shaojie Gong:** Resources, Conceptualization. **Yuxin Zhang:** Validation, Software, Formal analysis. **Xinyan Huang:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Asif Usmani:** Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A Supplementary material

Supplementary material to this article can be found online at https://doi.org/10.1016/j.undsp.2026.05.004.

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